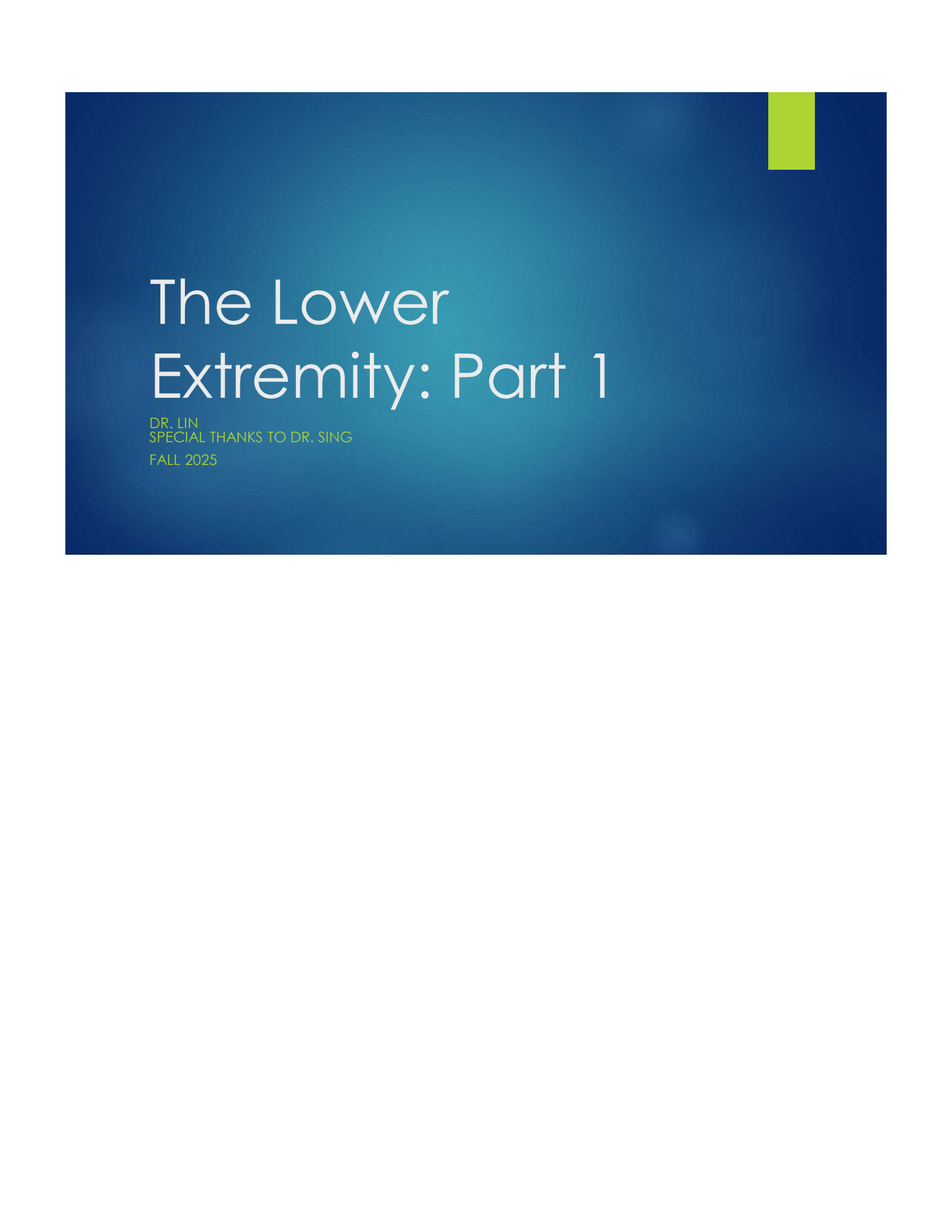




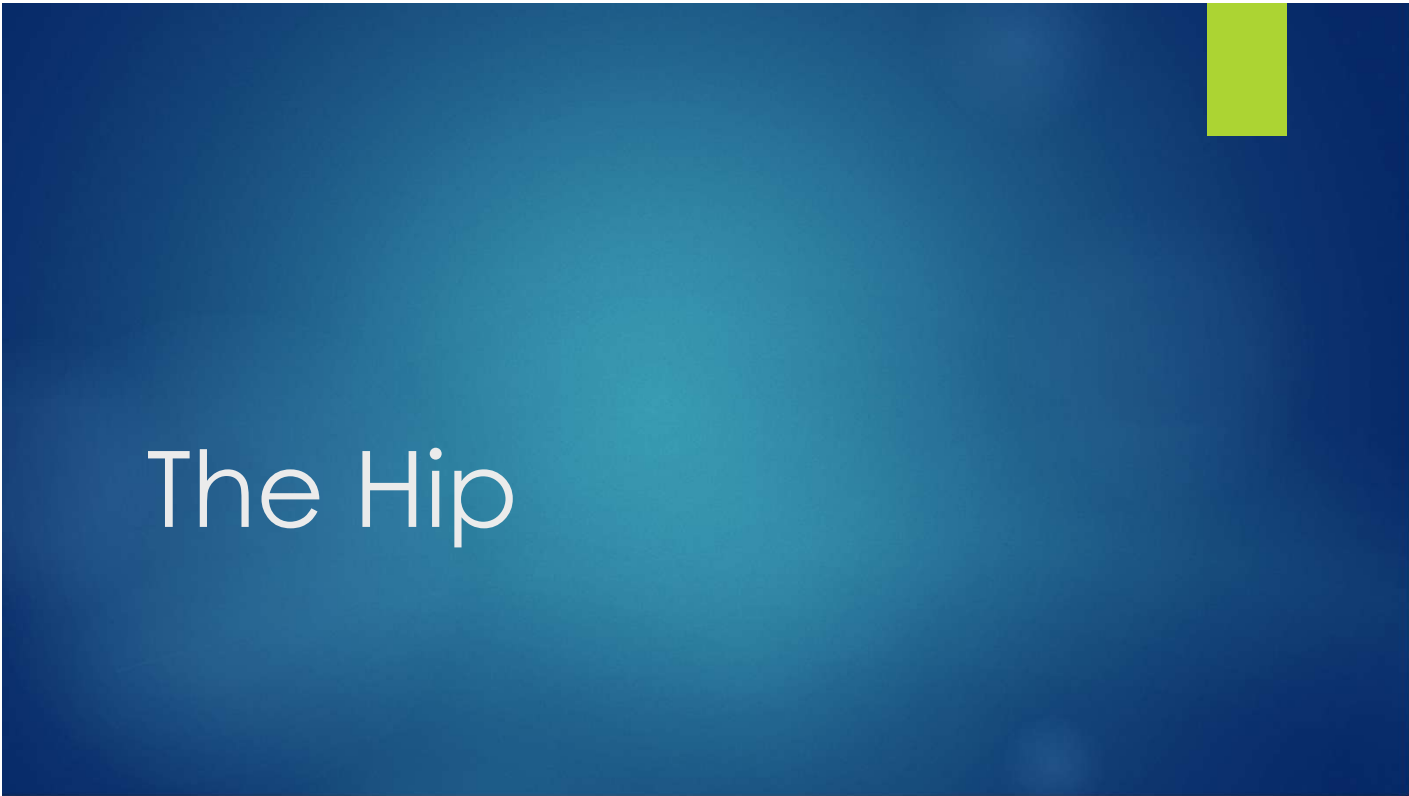
The Lower Extremity: Part 1

DR. LIN
SPECIAL THANKS TO DR. SING
FALL 2025

Objectives

For each of the **covered topics** in this lecture you should be able to:

- ▶ Recognize common presentations
- ▶ Understand and be able to describe the pathophysiology of the condition
- ▶ Know the essential elements of the physical exam and the accompanying anatomy
- ▶ List the diagnostic tests that may be necessary or helpful in making the diagnosis and differentiate from those that are not indicated
- ▶ Describe initial and non-surgical treatment and know when immediate surgery is indicated
- ▶ List and differentiate similarly presenting conditions



The Hip

Readings

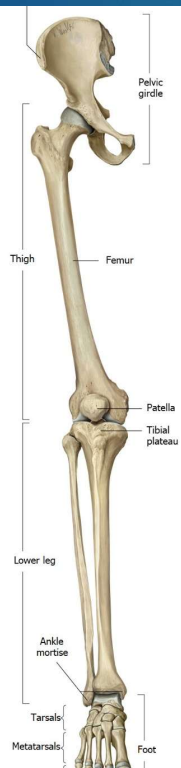
Essentials of Musculoskeletal Care, 6th ed

- ▶ General Principles of Hip
- ▶ Snapping Hip
- ▶ Strains of Hip
- ▶ Trochanteric Bursitis

Conditions of the Lower Extremities

part I : Pelvis and Hip

- ▶ Muscle Strains and Overuse Injuries
- ▶ Piriformis Syndrome
- ▶ Greater Trochanteric Bursitis
- ▶ Iliotibial Band Syndrome
- ▶ Snapping Hip
- ▶ Labral Tears of the Hip

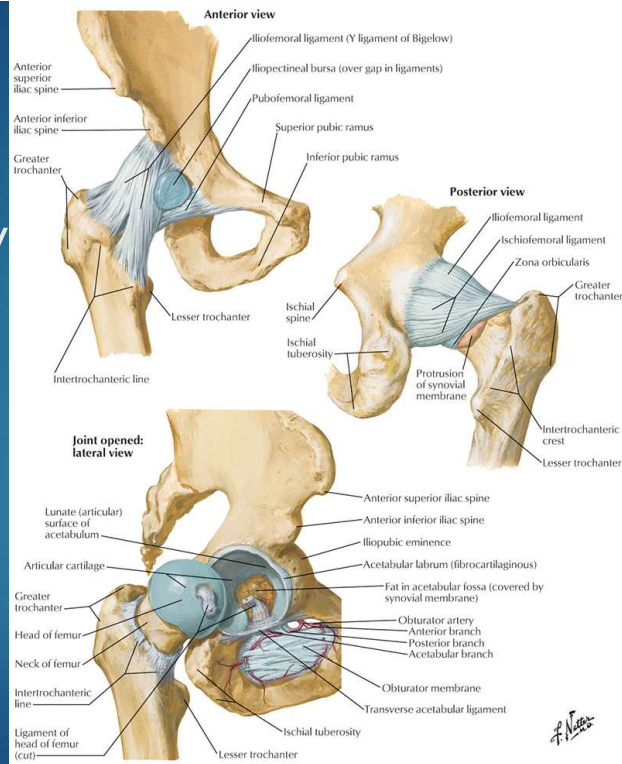


ANATOMY

Anatomy

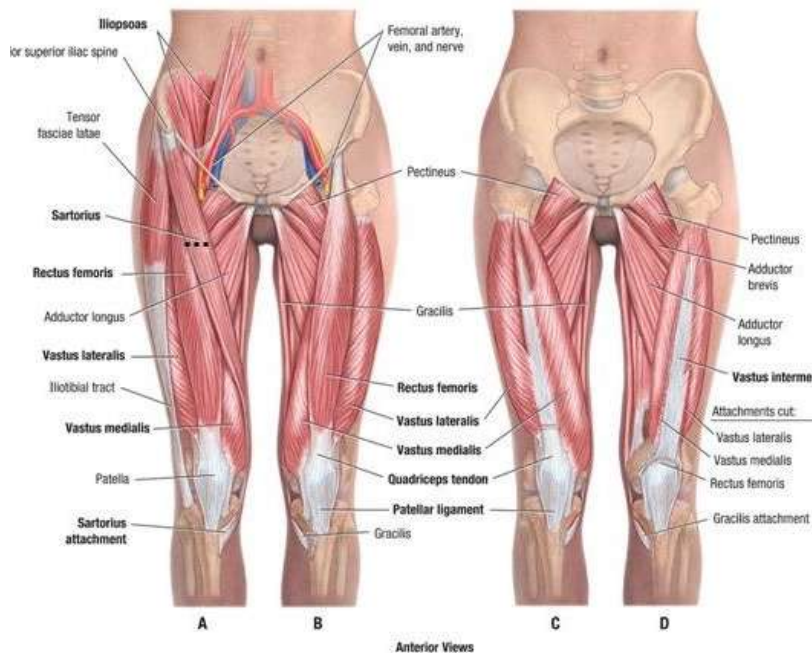
Ligaments

Muscles



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Ischium, illiums, pubis



Anatomy- Muscles

The tensor fasciae latae (TFL) is a muscle in the thigh that helps stabilize the hip and knee:

- **Location**

The TFL is a small, superficial muscle in the gluteal region of the lower limb, located on the outside of the hip. It's between the superficial and deep fibers of the iliotibial (IT) band, a thick band of connective tissue that runs down the outside of the leg to the knee.

- **Function**

The TFL works with the gluteus maximus, gluteus medius, and gluteus minimus to help with hip movements like flexion, abduction, and internal rotation. It also helps with knee flexion and lateral rotation.

- **Origin and insertion**

The TFL originates from the anterior iliac crest and attaches to the anterior superior iliac spine (ASIS). It inserts into the iliotibial tract, which attaches to the lateral condyle of the tibia

ITB:

- **Origin:** The ITB originates at the anterolateral iliac tubercle of the iliac crest.
- **Insertion:** The ITB inserts at Gerdy's tubercle on the lateral tibial plateau.
- **Function:** The ITB helps stabilize the knee, and the muscles associated with it help flex, extend, abduct,

and rotate the hip

So you say it's hip pain....

ANTERIOR PAIN

- ▶ Labral derangement
- ▶ Osteoarthritis
- ▶ Psoas spasm
- ▶ Avascular Necrosis of the hip
- ▶ Pelvic and Hip Fractures
- ▶ Anterior impingement

POSTERIOR PAIN

- ▶ Referred pain from low back
- ▶ SI joint pain
- ▶ Piriformis syndrome
- ▶ Gluteus medius bursitis
- ▶ Hamstring injury
- ▶ ITB Syndrome, snapping hip
- ▶ Greater Trochanteric bursitis

Remember not to forget red flags: what other emergency diagnosis that can present as back/hip/groin pain....

(AAA, nephrolithiasis, diverticulitis, PID, IBD)

Physical Exam

- ▶ Visual Exam
 - ▶ Evaluate for bruising, swelling, and erythema
- ▶ ROM
 - ▶ Flexion, Extension, Internal and External rotation, Abduction and Adduction
- ▶ Palpation
 - ▶ Major muscle groups and their bony origins
 - ▶ ITB band and G. Trochanter
- ▶ Gait
- ▶ Provocative Tests

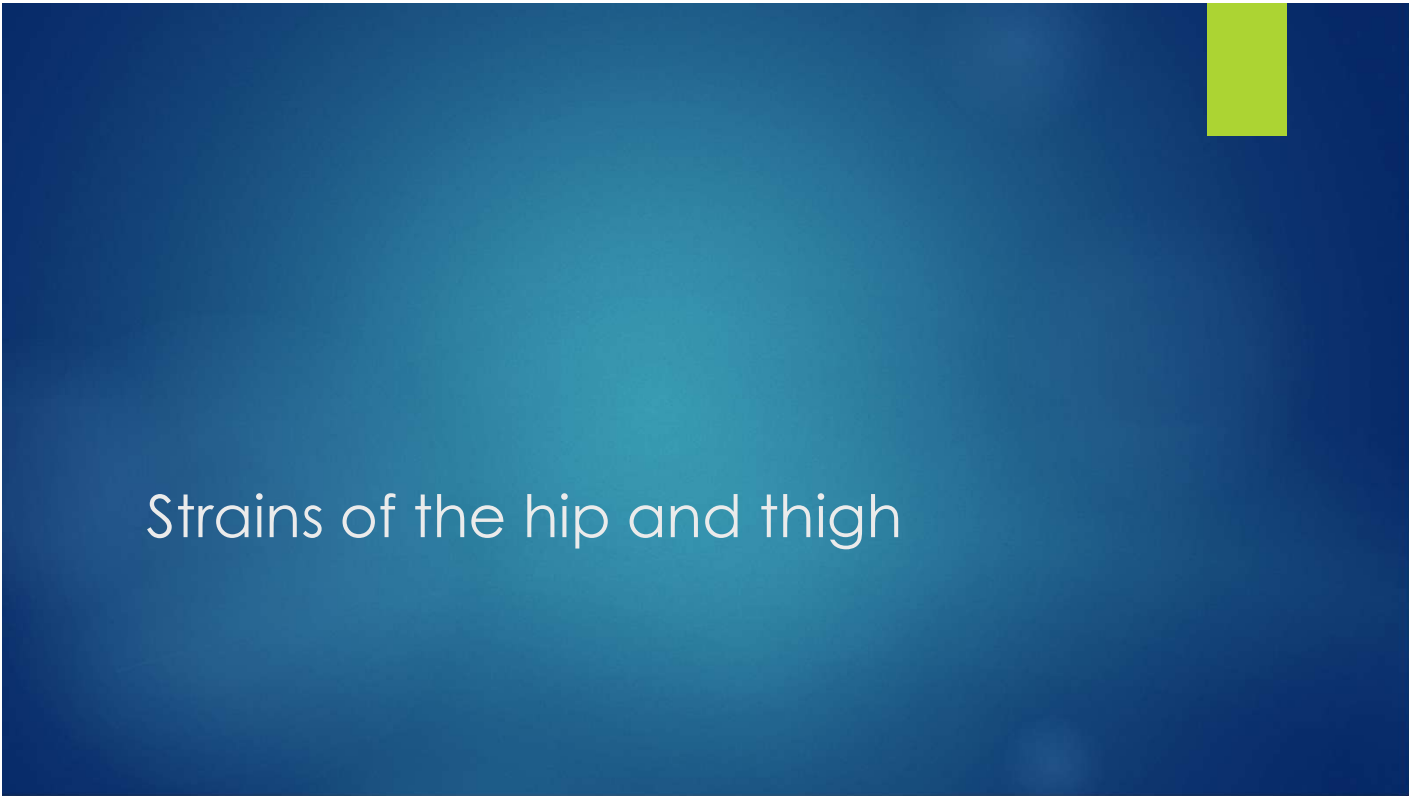
Common Causes for Hip Pain

- ▶ Anterior Hip Pain:
 - ▶ Labral derangement
 - ▶ Osteoarthritis
 - ▶ Psoas spasm
 - ▶ Avascular Necrosis of the hip
 - ▶ Pelvic and Hip Fractures
 - ▶ Anterior impingement
 - ▶ Slipped Capital Femoral Epiphysis

□Osteoarthritis will not be covered, but it is still important to note as a very common cause.

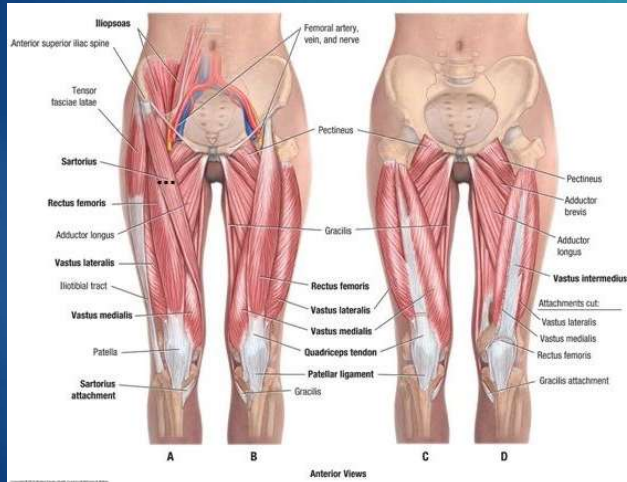
Common Causes for Hip Pain

- ▶ Posterior Hip Pain:
 - ▶ Referred pain from low back
 - ▶ SI joint pain
 - ▶ Piriformis syndrome
 - ▶ Gluteus medius bursitis
 - ▶ ITB Syndrome, snapping hip
 - ▶ Greater Trochanteric bursitis
 - ▶ Hamstring injury



Strains of the hip and thigh

Sprains/Strains



► Hip and thigh

► Hip adductors/abductors/rotators

- Adductor magnus, longus, brevis
- Gluteus maximus, medius, minimus
- Piriformis, Gemelli, Obturator externus

► Anterior/Posterior thigh muscles

- Quadriceps: Vastus lateralis, intermedius, medialis
- Hamstring

Adductor strain- intense repetitive side to side movement, skating, tenderness at tendon muscle junction

Quadriceps- rapid change of direction, pain with flexion

Hamstring- sprinting or hill running, MT junction, posterior ischial tuberosity

Sprain: Ligament

Strain: muscle/tendon



Pathophysiology

- ▶ Most common cause of hip pain in adults
- ▶ Consider abdominals, hip flexors (iliopsoas, sartorius, and rectus femoris) and hip adductors
- ▶ Frequently happens with vigorous muscle contraction while the muscle is on stretch
 - ▶ Ex: soccer player forcibly flexes the hip to kick a ball and the leg is blocked by an opponent

History is important, what are they doing physically? Job? Lifting?

Remember Hip pain that does not seem to improve could be due to lumbar radiculopathy and require MRI and Orthopedic evaluation.

Differential

- ▶ Hip avulsion fractures (may be seen with strains)
- ▶ Osteonecrosis of the hip (chronic dull ache in the groin, inner thigh or buttock)
- ▶ Pelvic or proximal femoral tumors (pain at rest or at night, increased pain with weight bearing)

Presentation & Physical Exam

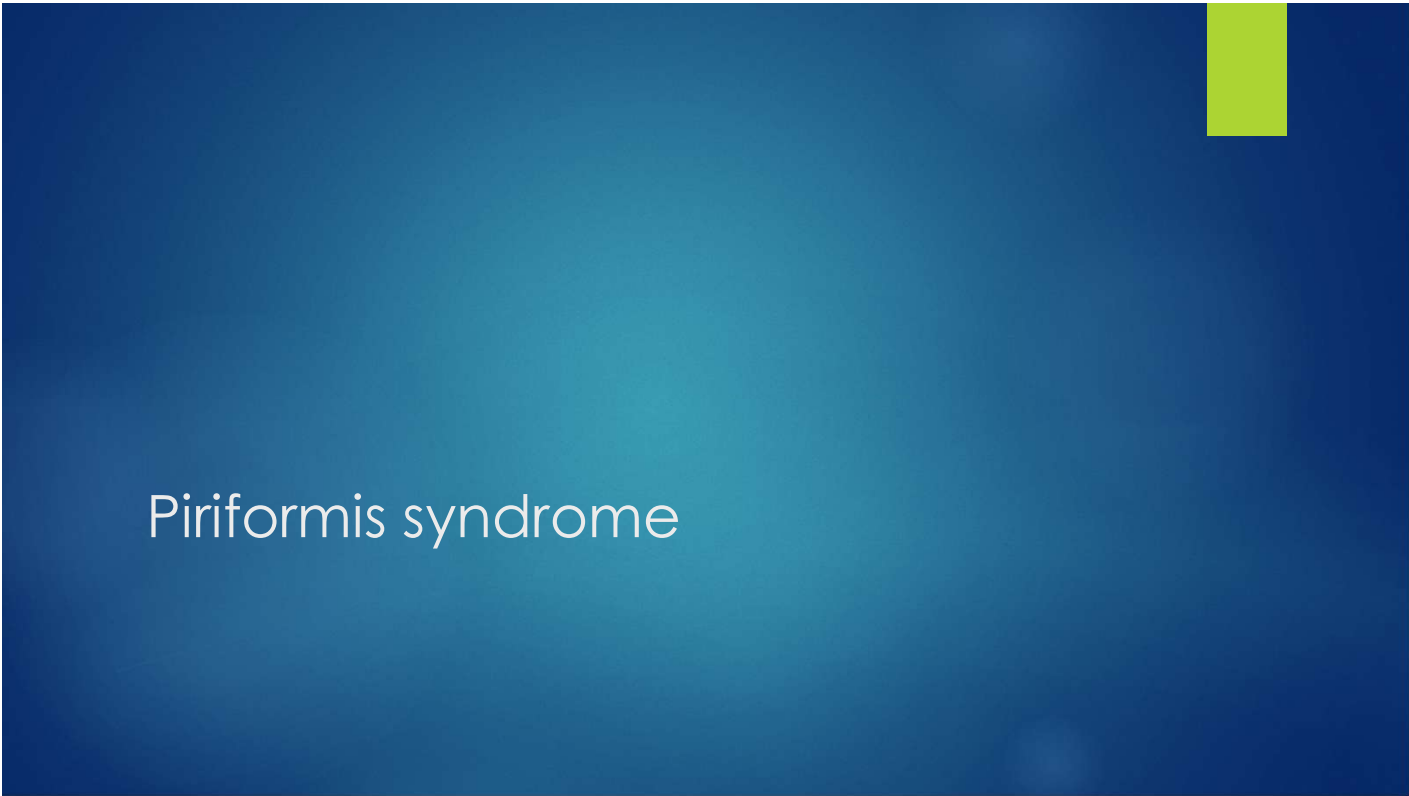
- ▶ Presents as pain over the injured muscle that is exacerbated when that area continues to be used during strenuous activities
- ▶ Tenderness to palpation
- ▶ Increased pain with passive motion
- ▶ Pain with resistive strength testing
- ▶ Ecchymosis dependent on level of strain

Diagnosis

- ▶ Xray can rule out fracture if suspected
- ▶ Diagnosis is clinical

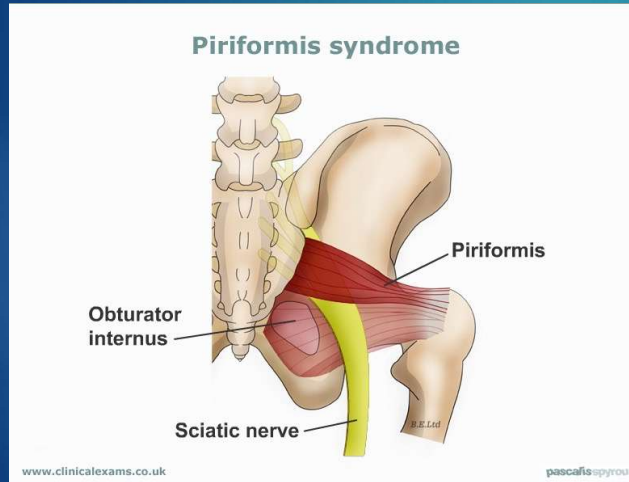
Treatment

- ▶ RICE Therapy (Rest, Ice, Compression, Elevation)
- ▶ Ice/Heat and NSAIDs can help
- ▶ Specific exercises
- ▶ Physical therapy



Piriformis syndrome

Pathophysiology



- ▶ Sciatic nerve entrapment that occurs anterior to piriformis muscle or posterior to obturator internus at the level of ischial tuberosity

In >80% of the population, the sciatic nerve courses deep to and exits inferiorly to the piriformis muscle belly/tendon

Differential

- ▶ Lumbar radiculopathy
- ▶ Primary sacral dysfunction
- ▶ Innominate dysfunction

Presentation & Physical Exam

- ▶ Presents as pain in the posterior gluteal region and migrating down the back of the leg
 - ▶ Pain is burning or aching in nature
- ▶ Often a source for sciatica
- ▶ On PE, FAIR test can reproduce symptoms
 - ▶ Flexion, Adduction, and Internal Rotation of the hip



*

Diagnosis

- ▶ Clinical
- ▶ No imaging is needed

Treatment

- ▶ Rest
- ▶ OMM
- ▶ NSAIDs
- ▶ Muscle relaxants
- ▶ Physical therapy
- ▶ Steroid injections can be considered

Trigger point txt, release sacrum, treating pelvic strains with OMM can be very helpful!



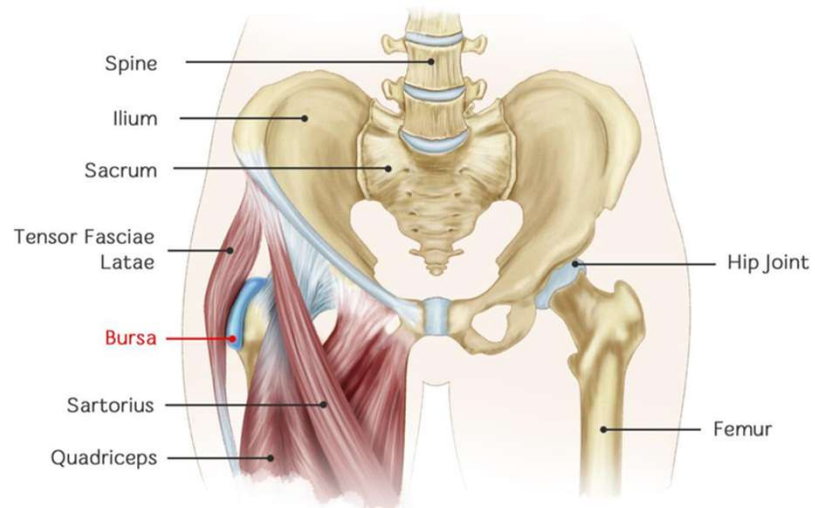


Trochanteric bursitis

involves inflammation of the greater trochanteric bursa, which is located between the greater trochanter and the iliotibial band.

Trochanteric Bursitis

REHAB
My Patient.com



Pathophysiology

- ▶ Repetitive trauma caused by iliotibial band tracking over trochanteric bursa
- ▶ This can irritate the bursa causing inflammation
- ▶ Common in female runners
- ▶ Associated with training on banked surfaces
- ▶ Relates to IT Band syndrome that causes lateral knee pain and lateral thigh pain



History is important, physical activity, job (bending, lifting).

Differential

- ▶ Osteoarthritis of the hip
- ▶ Sciatica
- ▶ Septic arthritis of the hip
- ▶ Snapping hip
- ▶ Trochanteric fracture
- ▶ Metastatic tumor

Presentation & Physical Exam

- ▶ Lateral sided hip pain
 - ▶ Pain may radiate to knee or buttock
 - ▶ Pain is worse when pt raises from a seated position then lessens after a couple steps
 - ▶ Pain increases after walking for 30 min or more
 - ▶ Pain often reported at night as pt is unable to lie on affected side
- ▶ Hip joint is not involved
- ▶ On PE, **pain with palpation** over greater trochanter

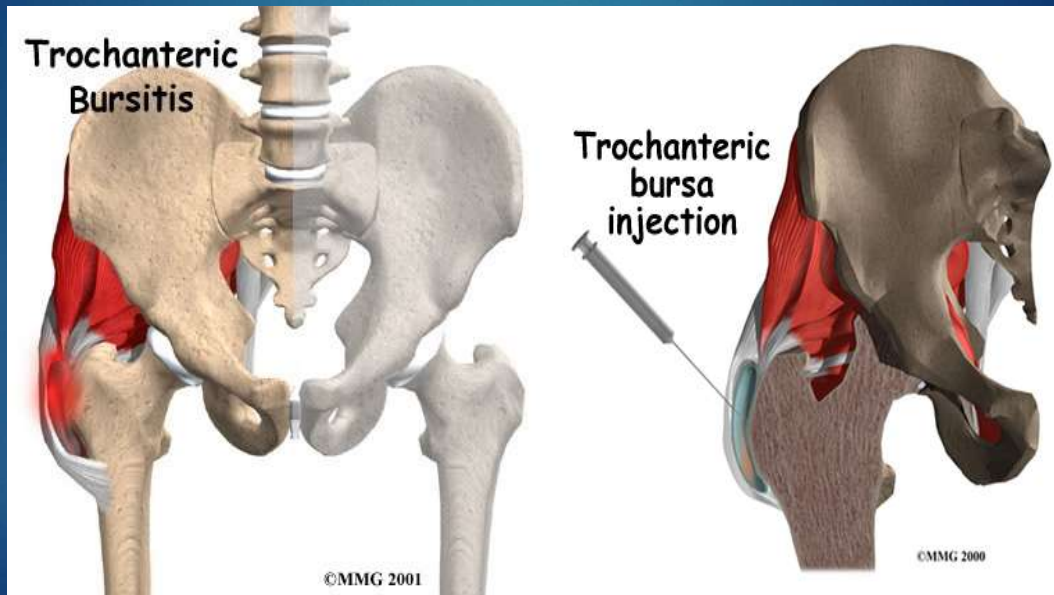
Diagnosis

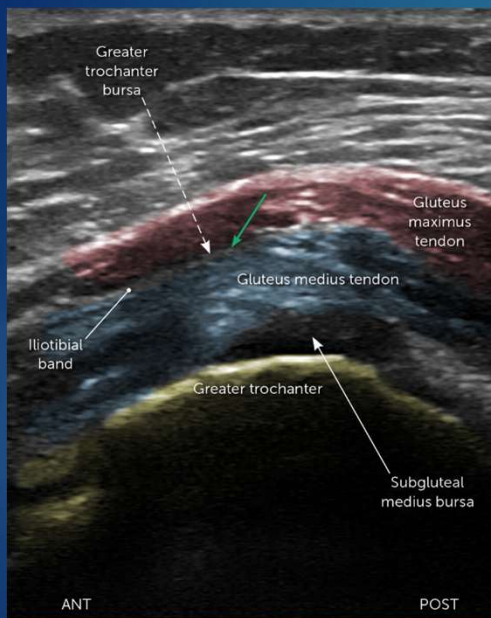
- ▶ X-rays necessary to rule out bone abnormalities

Treatment

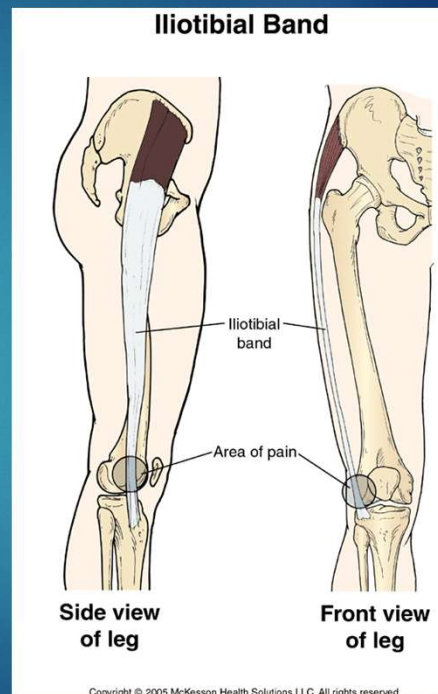
- ▶ NSAIDs
- ▶ Activity modification
- ▶ IT Band stretching
- ▶ Ice
- ▶ Injection

Trochanteric Bursitis





ITB Syndrome



*Look up more&

ITB Syndrome

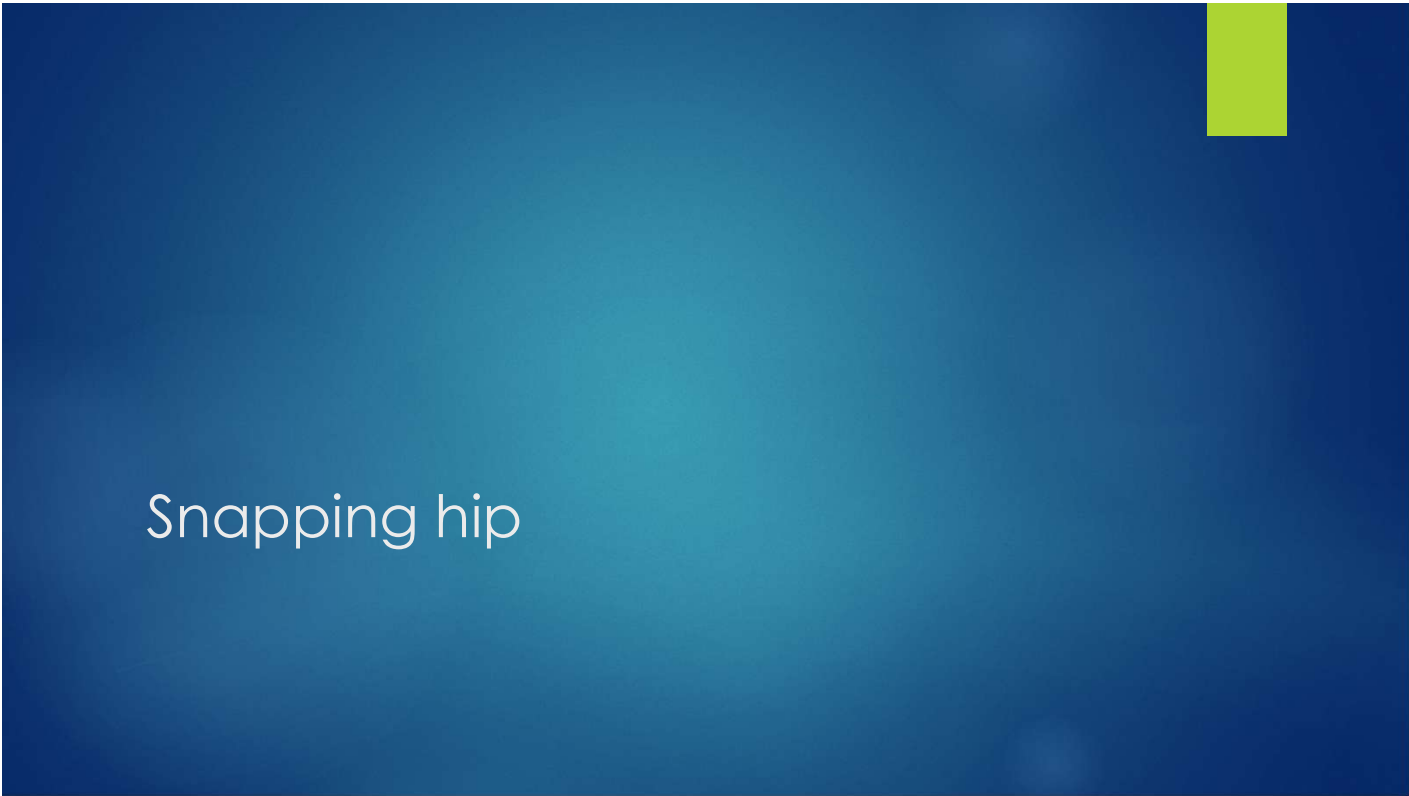
- ▶ Very common cause of pain in hip
- ▶ Common with running, cycling, and walking
- ▶ May be caused by tight ITB or adhesions to ITB following trauma
- ▶ Diagnosed with palpation over IT band and Ober's test
- ▶ Improved with Ice/heat/stretching/roller

Ober's Test



Roller for ITB Syndrome

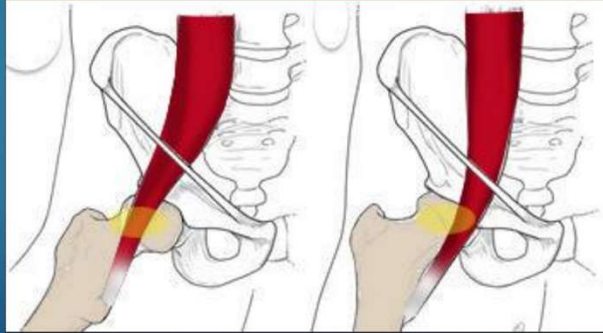




Snapping hip

Pathophysiology

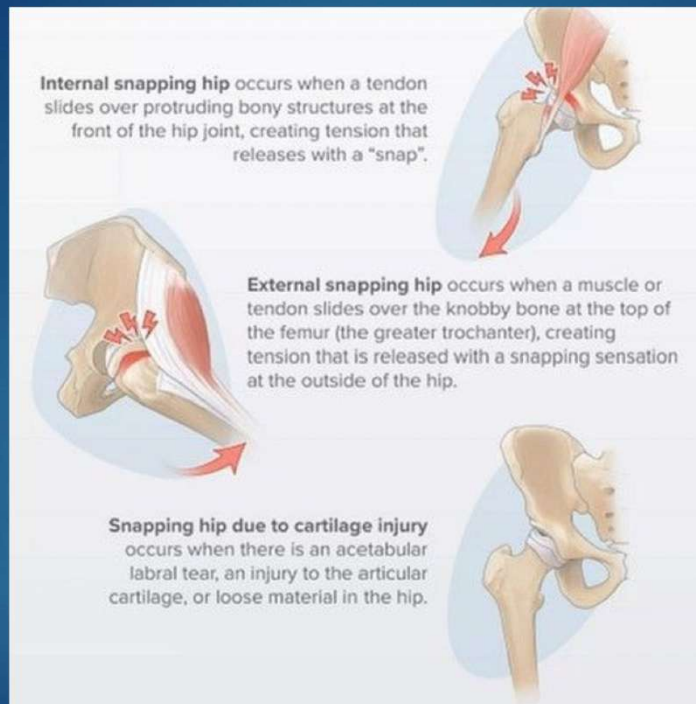
- ▶ Condition characterized by a snapping sensation in the hip
 - ▶ Caused by motion of muscles and tendons over bony structures around the hip joint
- ▶ Common in athletes and dancers in their teens and twenties



Pathophysiology

- ▶ 3 types of snapping hip exist with different causes
 - ▶ **External snapping hip**
 - ▶ Caused by iliotibial tract sliding over greater trochanter
 - ▶ **Internal snapping hip**
 - ▶ Most common form
 - ▶ Caused by iliopsoas tendon sliding over: femoral head, prominent iliopectineal ridge, exostoses of lesser trochanter, or iliopsoas bursa
 - ▶ **Intra-articular snapping hip**
 - ▶ Caused by loose bodies in the hip or labral tears

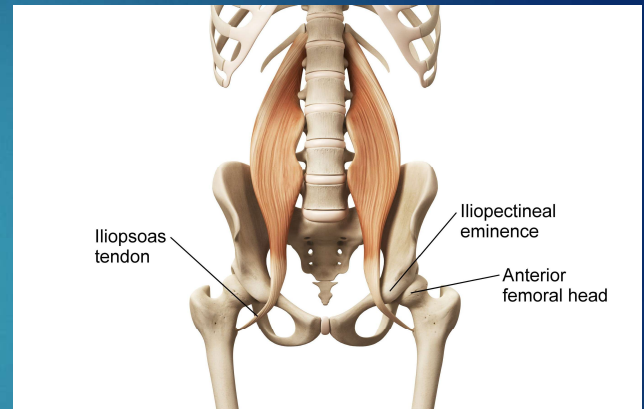
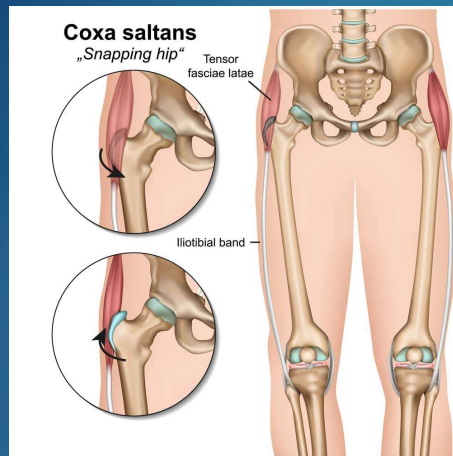
A loose body is a small piece of bone or cartilage that breaks off and floats freely in a joint, causing pain, swelling, and limited range of motion



External snapping hip is most commonly attributed to the iliotibial band moving over the greater trochanter of the femoral head during hip movements in flexion, extension, and external or internal rotation.

Internal snapping hip most commonly occurs as the iliopsoas tendon snaps over underlying bony prominences, such as the iliopectinal eminence or the anterior aspect of the femoral head.

https://www.youtube.com/watch?v=-W3o_WADI1M



Differential

- ▶ Osteoarthritis
- ▶ Osteonecrosis of the femoral head
- ▶ Tear of acetabular labrum
- ▶ Osteochondral loose body (fragment of bone and cartilage within the joint)

Presentation & Physical Exam

- ▶ Presents as a snapping sensation in and around hip joint
 - ▶ May be painful or painless
 - ▶ Patient is often able to reproduce snapping
 - ▶ Aggravated by activity
- ▶ PE:
 - ▶ External snapping hip is often visible
 - ▶ Internal snapping hip may be audible

<https://healthy.kaiserpermanente.org/health-wellness/health-encyclopedia/he.snapping-hip-syndrome-exercises.bo1646>

Diagnosis

- ▶ Xrays are usually normal but can rule out other disorders, foreign body, etc
- ▶ Ultrasound may demonstrate the snapping band
- ▶ Usually a Benign condition with external and internal type, most patients are concerned about the snap but do not have pain.
- ▶ If you suspect a loose body or labral tear, more work up is needed

Ultrasound show?



► J Ultrason. 2016 Sep 7;16(66):296–303. doi: [10.15557/JoU.2016.0030](https://doi.org/10.15557/JoU.2016.0030)

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Internal snapping hip syndrome in dynamic ultrasonography

[Małgorzata Piechota](#)^{1,✉}, [Jarosław Maczuch](#)¹, [Jarosław Skupiński](#)¹, [Karolina Kukawska-Sysio](#)¹, [Wojciech Wawrzynek](#)¹

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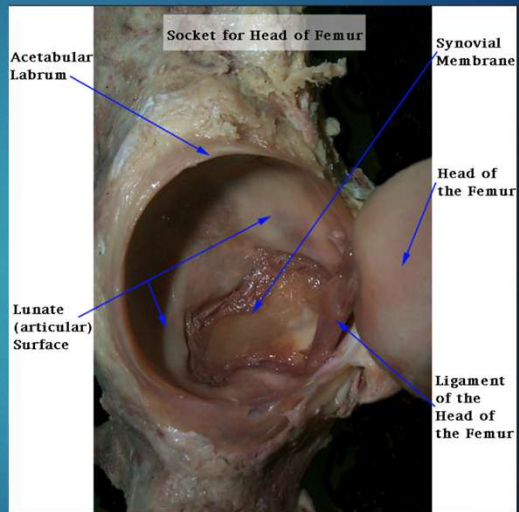
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Treatment

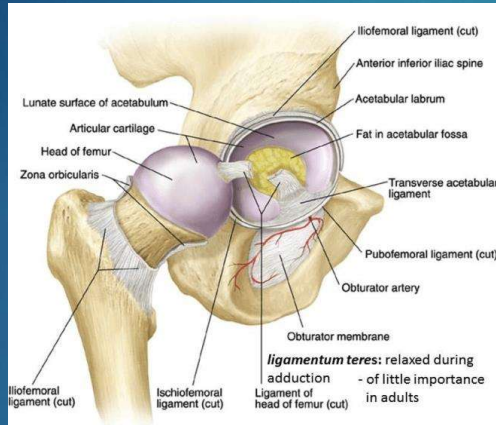
- ▶ Requires no treatment
- ▶ Activity modification
- ▶ Physical therapy
- ▶ Surgery can be indicated if painful and failure of conservative treatment

For external snapping hip syndrome, loosening of the iliotibial band is usually the goal and can be accomplished with either open or arthroscopic procedures. For internal snapping hip syndrome, open or arthroscopic procedures also are available to lengthen or release the iliopsoas tendon.

Labral tear



Pathophysiology



- ▶ Traumatic tear of the acetabular labrum that may lead to pain or intra-articular snapping hip
- ▶ Seen in all age groups and commonly in active females
- ▶ Happens due to hip dysplasia, trauma, capsular laxity, or joint degeneration

Differential

- ▶ Osteoarthritis
- ▶ Septic arthritis
- ▶ Piriformis syndrome
- ▶ Snapping hip
- ▶ Bursitis

Presentation & Physical Exam

- ▶ Presents as mechanical hip pain and snapping
 - ▶ May be associated with a sensation of locking, usually positional and painful
- ▶ PE:
 - ▶ Anterior labral tear: Pain if hip is brought from a fully flexed, externally rotated, and abducted position to a position of adduction, internal rotation and extension.
 - ▶ Posterior labral tear: Pain if hip is brought from a flexed, adducted and internally rotated position to one of abduction, external rotation and extension
- ▶ Fitzgeralds Test
<https://www.youtube.com/watch?app=desktop&v=FyXW-ExBiOM>

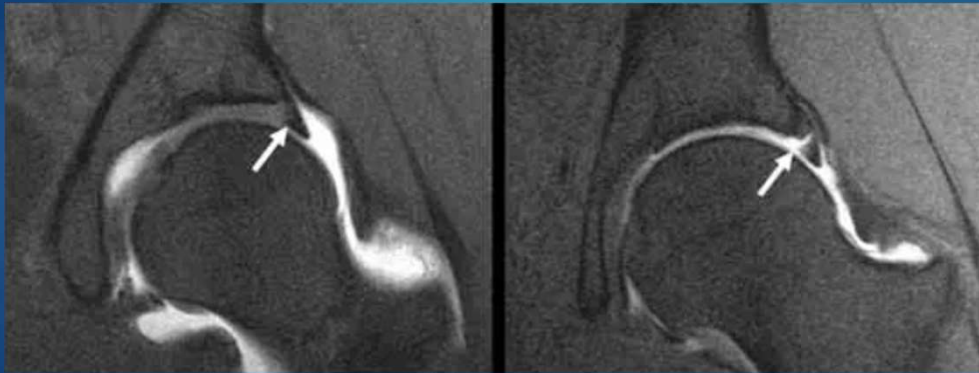


Flexion- sharp groin pain that increases with activity that load hip;

<https://www.youtube.com/watch?app=desktop&v=FyXW-ExBiOM>

Diagnosis

- ▶ MRI arthrogram is the imaging study of choice
 - ▶ 92% sensitive

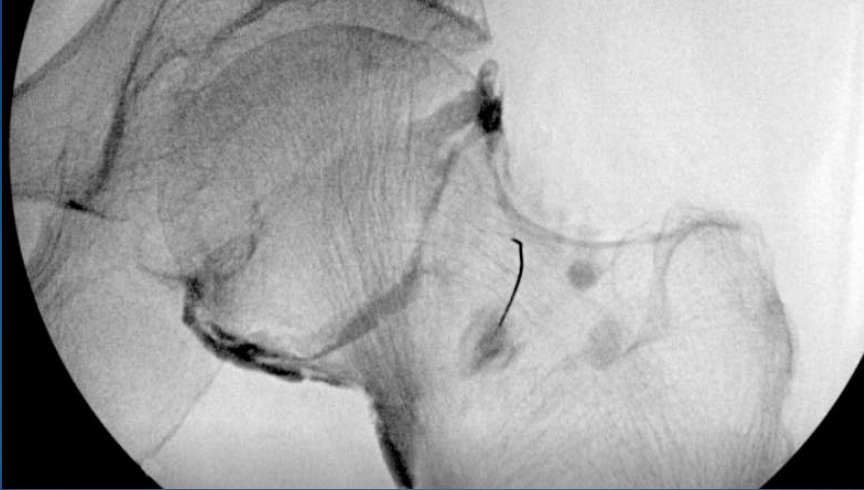


MRI scan of a normal acetabular labrum (left) and a torn labrum (right).

Good to distinguish from impingement.

Treatment

- ▶ Rest, NSAIDs, PT, steroid injections initially
- ▶ Arthroscopic labral debridement
 - ▶ Indicated if nonoperative modalities fail
 - ▶ Removes unstable portion of labrum



Non traumatic Avascular Necrosis (AVN) of the Femoral Head

- ▶ Thought to be due to cause of decreased blood flow to bone leading to necrosis
 - ▶ Seen in adults
 - ▶ Trauma
 - ▶ Steroid use, chemotherapy
 - ▶ Co-morbidity with etoh use, SLE, blood coagulation disorders
- ▶ Presentation:
 - ▶ Subacute, the beginning of necrosis does not create pain
 - ▶ Groin pain, then thigh and buttock
 - ▶ Pain with motion, weight bearing, but at night and rest pain become persistent
 - ▶ Finally ROM and limp in later stages

Certain anatomic sites are prone to osteonecrosis and include the femoral head, humeral head, knee, and certain bones of the foot (talus) and wrist (scaphoid), lunate (Kienbock's dz) but any bone can be involved.

Decreased collateral circulation

AVN

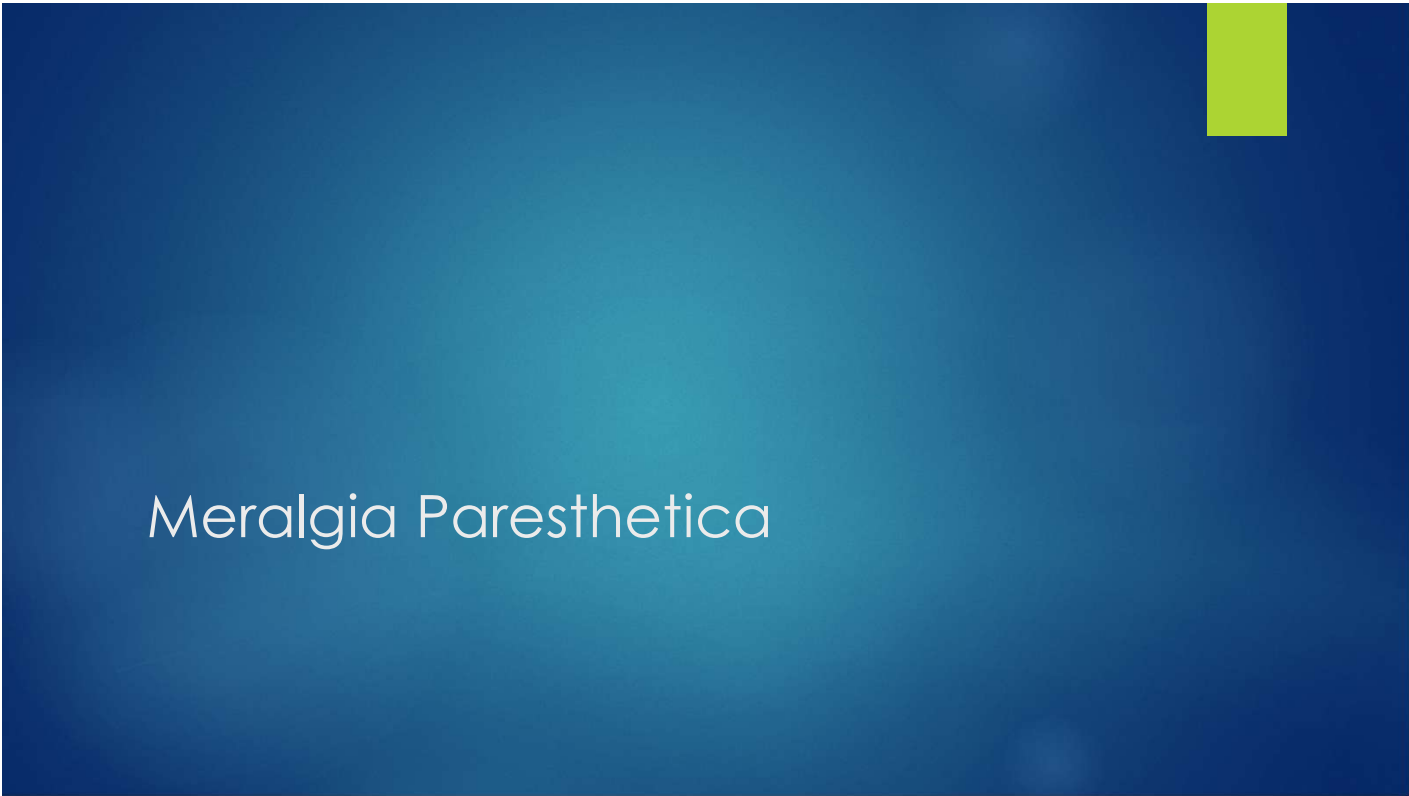
- ▶ Diagnosis
 - ▶ X Rays, first consider bilateral
 - ▶ MRI has a reported sensitivity of up to 100 percent for the diagnosis of avascular necrosis
 - ▶ Changes can be seen early in the course of disease when radiographs or other studies are negative
- ▶ Management
 - ▶ Maximize function
 - ▶ Progressive disease eventually will require surgical intervention w Hip replacement



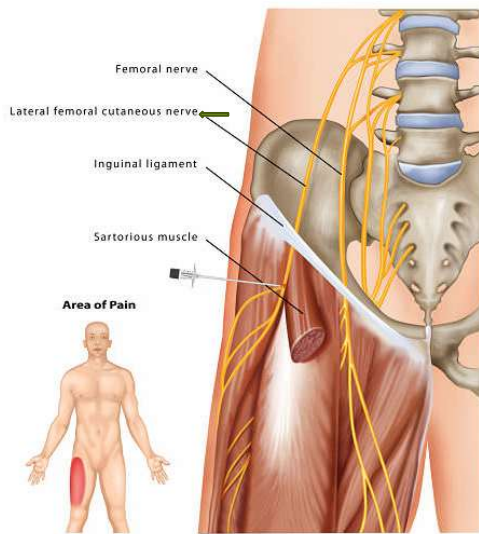
Level of bone involvement will stage disease, prevent collapse.

AP X-ray of pelvis. Example of a patient whose X-ray shows a completely destroyed hip compared to a nearly normal hip on the other side. This person was referred for a hip replacement.





Meralgia Paresthetica



Meralgia Paresthetica

- ▶ Clinical syndrome of pain, dysesthesia, or both in the anterolateral thigh associated with compression of the lateral femoral cutaneous nerve
- ▶ associated with advancing age, diabetes, and obesity.
- ▶ due to restrictive clothing, body habitus, or other factors that result in local compression of the nerve at the inguinal ligament.

It's caused by compression of the lateral femoral cutaneous nerve

Meralgia Paresthetica

- ▶ Diagnosis of meralgia paresthetica is based primarily upon:
 - ▶ unique description of pain and paresthesias in the characteristic location
 - ▶ Isolated sensory abnormalities on examination
 - ▶ absence of neurologic abnormalities of the lower leg.
- ▶ Treatment:
 - ▶ Remove compressive force or weight
 - ▶ Self-limited, benign disease in most patients.
 - ▶ More than 90 percent of patients respond to conservative measures alone, although recurrent symptoms are common





<https://www.youtube.com/watch?v=OylFThA8YSs>

Condition	Pathophysiology	Key Clinical Features	Diagnosis	Treatment
Hip Strains	Overstretching or vigorous contraction of hip muscles	Localized pain over injured muscle, pain with motion	Clinical; X-ray to rule out fracture	RICE, NSAIDs, PT
Piriformis Syndrome	Sciatic nerve entrapment by piriformis muscle	Gluteal pain radiating down leg; positive FAIR test	Clinical	Rest, NSAIDs, PT, muscle relaxants
Trochanteric Bursitis	IT band rubbing over trochanteric bursa	Lateral hip pain, worse when rising or lying on side	X-ray to rule out bony pathology	NSAIDs, activity modification, IT band stretching, injections
IT Band Syndrome	IT band tightness or adhesions	Lateral thigh/knee pain; positive Ober's test	Clinical	Ice, heat, stretching, roller therapy
Snapping Hip Syndrome	Tendons snapping over hip structures	Snapping sensation in hip, may be audible	X-ray to rule out other pathology; U/S	Activity modification, PT; surgery if symptomatic
Labral Tear	Tear of acetabular labrum	Mechanical hip pain, snapping, possible locking	MRI arthrogram	NSAIDs, PT, arthroscopic debridement if persistent
Avascular Necrosis (AVN)	Reduced blood flow to femoral head	Groin pain progressing to thigh and buttock pain	X-ray, MRI for early detection	Maximize function, eventual hip replacement if severe
Meralgia Paresthetica	Compression of lateral femoral cutaneous nerve	Anterolateral thigh pain/paresthesia; sensor ↓ עור	Clinical, based on characteristic location	Remove compression, conservative care



Hip Cases



Case 1

- ▶ A 35 year old male presents with pain in his hip. He states the pain is burning in nature and radiates down the back of his leg. On exam, when asked to point to the pain, he points to the gluteal region. FAIR (flexion, adduction, and internal rotation of the hip) can reproduce his pain.

Case 1

- ▶ What is your working diagnosis?
- ▶ What nerve is involved?
- ▶ What muscles lead to entrapment?
- ▶ What imaging is needed to make the diagnosis?
- ▶ What treatment is first line?

Piriformis syndrome

Sciatic nerve

Anterior to piriformis muscle or posterior to obturator internus at the level of the ischial tuberosity

None, Diagnosis is Clinical

RICE, PT, OMM

<https://www.degruyter.com/document/doi/10.7556/jaoa.2008.108.11.657/html?lang=en>

Case 2

- ▶ A 19 year old female ballet dancer presents to your office with left hip pain. She states she feels a snapping sensation that is worse when she dances. She states she can hear a snap when she moves in a certain way and is able to demonstrate that in your office.

Case 2

- ▶ What is your working diagnosis?
- ▶ What anatomic structures are involved in this condition?
- ▶ What imaging is needed?
- ▶ What is the treatment for this condition?

Internal snapping hip

Iliopsoas sliding over: femoral head, prominent iliopectineal ridge, exostoses of lesser trochanter, or iliopsoas bursa

X-ray or US should be done to rule out loose bodies in the hip

Benign condition. Activity modification or PT can be used.

Case 3

- ▶ A 48 year old female with a past medical history of breast cancer presents to your office with right hip pain. She is a runner and has recently started training for a marathon. She states the pain starts after she runs about 3 miles. She also has pain when she gets up from a chair. She sometimes has pain at night. On exam, she points to the lateral right hip as the source of her pain. You can elicit pain with palpation over the area.

Case 3

- ▶ What is your working diagnosis?
- ▶ What imaging is needed?
- ▶ What special test should be done in this case?
- ▶ What treatment is indicated?
- ▶ What patient education is needed?

Greater trochanteric bursitis

X-ray is needed to rule out malignancy. She has a history of breast CA and night pain so this is a red flag that indicates imaging is necessary. Otherwise, diagnosis is clinical

Ober's test can test for IT Band tightness

NSAIDs, ice, injection, consider foot orthoses/correct limb length discrepancy.

Activity modification is key. Roller for ITB syndrome, making sure she has good shoes, changing the surface she is running on and decreased millage temporarily.

Case 4

- ▶ A 22 year old female presents to your office with right hip pain. She states she is a collegiate golfer and the pain has been getting worse throughout golf season. She denies injury to her hip. She does state she feels a "locking" at times. On exam, she points to the anterior hip as the source of her pain. She has pain if she is moved from the FABER test into hip extension, internal rotation and adduction.

Case 4

- ▶ What is your working diagnosis?
- ▶ What is the imaging study of choice to make this diagnosis?
- ▶ What is first line treatment?
- ▶ What treatment is indicated if first line treatment fails?

Anterior labral tear of the hip

MRI arthrogram

RICE, NSAIDs, PT

Arthroscopic labral debridement

Relax...you go this!

